



SIMATS ENGINEERING



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ASSESSMENT TOOL -2

CO4_AT2

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Course code : CSA6502

Course name: Generative Ai And Large Scale Models

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Question 1 :-

Compare the architectural strengths and limitations of BERT, GPT, T5, and LLaMA for language understanding and language generation.

ANSWER :-

Introduction

Natural Language Processing (NLP) has experienced remarkable progress with the introduction of Transformer-based models. Before Transformers, recurrent neural networks (RNNs) and long short-term memory (LSTM) networks were widely used but struggled with long-range dependencies and parallel processing. The Transformer architecture, introduced by Vaswani et al. (2017), revolutionized NLP through the self-attention mechanism, enabling models to process entire sequences efficiently.

Among the most influential Transformer-based models are BERT (Bidirectional Encoder Representations from Transformers), GPT (Generative Pre-trained Transformer), T5 (Text-to-Text Transfer Transformer), and LLaMA (Large Language Model Meta AI). Each model has a unique architecture and is optimized for different NLP tasks such as language understanding, text generation, translation, summarization, and question answering.

This assignment compares the architectural design, strengths, limitations, and applications of these four models.

1. BERT (Bidirectional Encoder Representations from Transformers)

Architecture

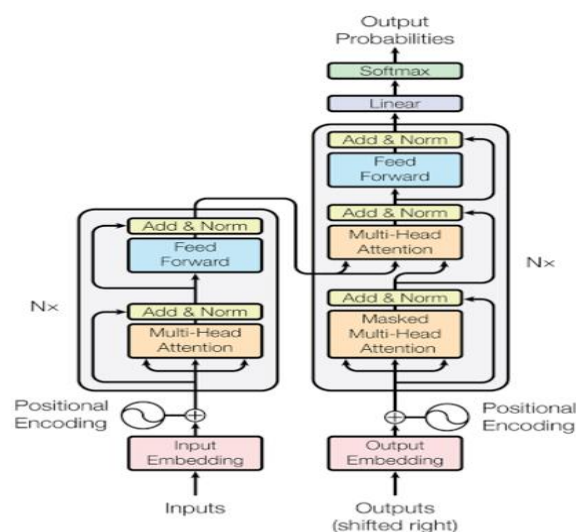


Fig. 2- BERT Model Architecture

Training Objectives

- Masked Language Modeling (MLM)
- Next Sentence Prediction (NSP)

The MLM task randomly masks words in a sentence and trains the model to predict them, enabling deep contextual understanding.

Strengths

- Excellent contextual understanding because it considers both left and right contexts.
- Achieves high accuracy in language understanding tasks.
- Performs exceptionally well in:
 - Sentiment Analysis
 - Named Entity Recognition (NER)
 - Question Answering
 - Text Classification
 - Document Classification
- Produces high-quality contextual word embeddings.
- Requires relatively smaller datasets for fine-tuning.

Limitations

- Cannot naturally generate text because it is encoder-only.
- Requires separate fine-tuning for different tasks.
- Computationally expensive during pre-training.
- Limited capability in conversational AI and long-form content generation.

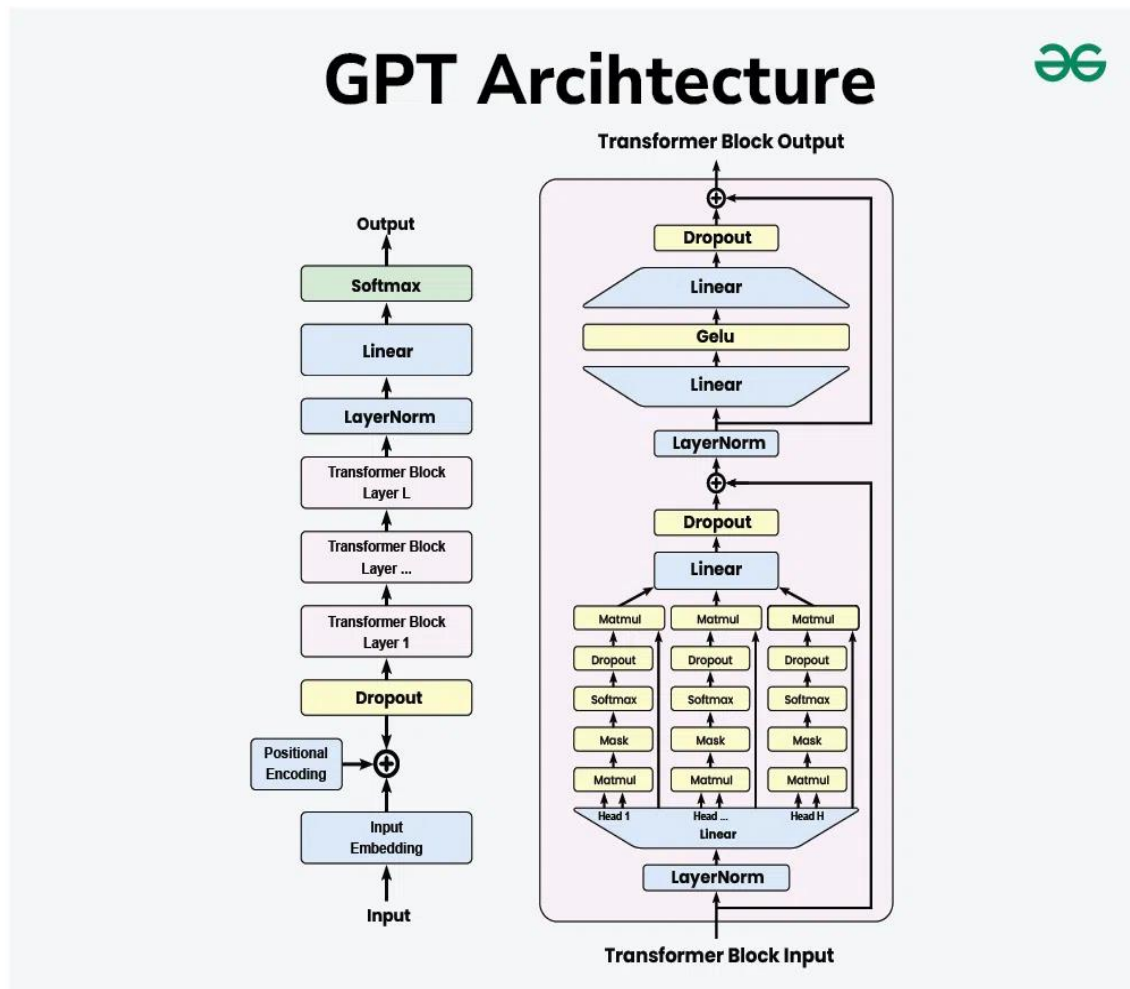
Applications

- Search engines
- Spam detection
- Medical text classification

2. GPT (Generative Pre-trained Transformer)

Architecture

GPT, developed by OpenAI, is built using the Transformer Decoder architecture. It predicts the next word in a sequence using causal (autoregressive) attention, generating text one token at a time.



Training Objective

- Next Token Prediction

The model learns language patterns by predicting the next word based on previously generated words.

Strengths

- Excellent text generation capability.

- Produces coherent and human-like responses.
- Performs well with:
 - Chatbots
 - Content writing
 - Story generation
 - Code generation
 - Email drafting
- Supports Zero-shot, One-shot, and Few-shot learning.
- Handles long conversations effectively.

Limitations

- Cannot fully utilize future context during training because it is unidirectional.
- May generate incorrect or fabricated information (hallucinations).
- Requires significant computational resources.
- Sometimes produces repetitive or biased outputs.

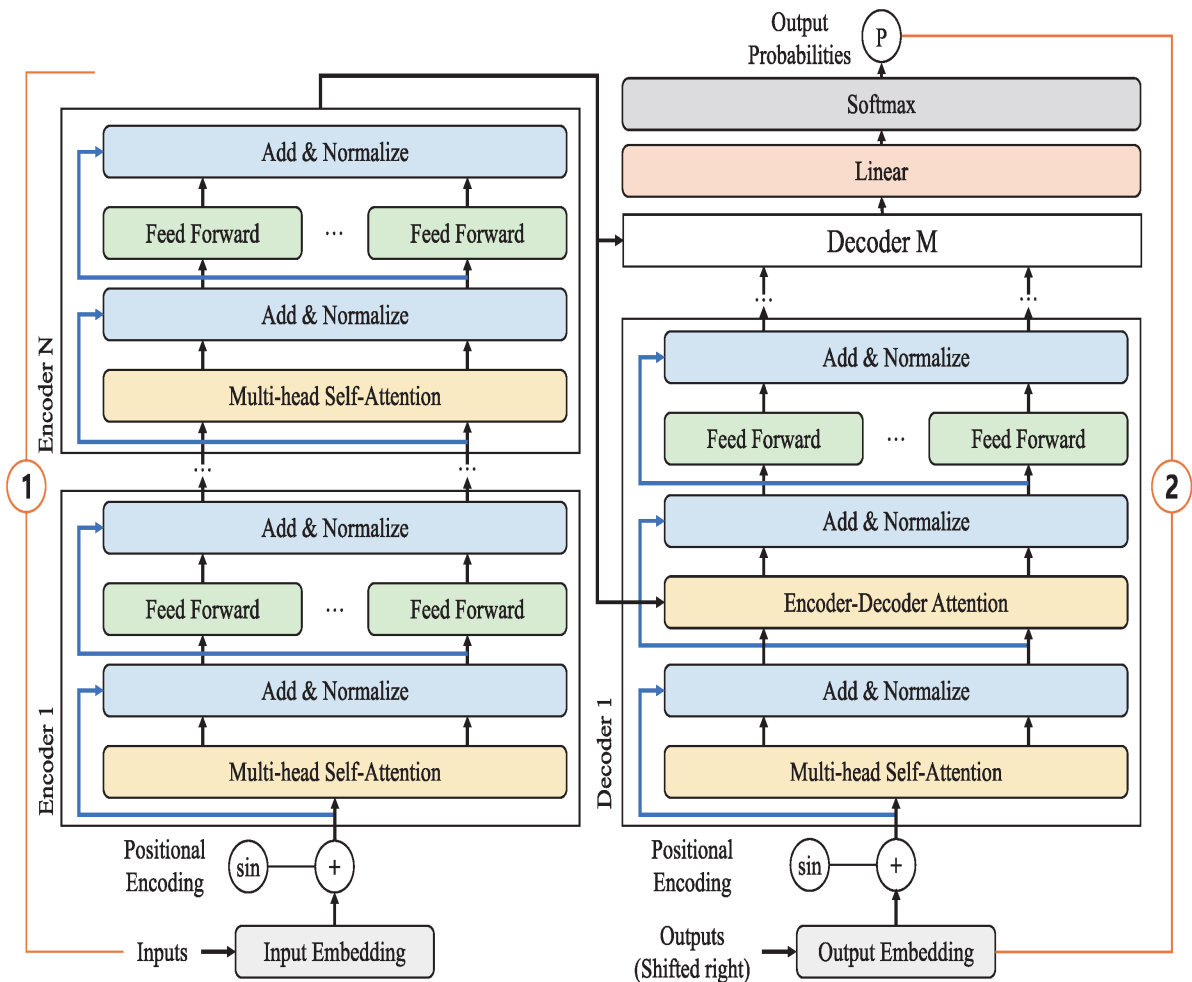
Applications

- Virtual assistants
- AI chatbots
- Content generation
- Programming assistants
- Translation
- Text summarization

3. T5 (Text-to-Text Transfer Transformer)

Architecture

T5, developed by Google Research, uses the Encoder-Decoder Transformer architecture. Every NLP task is converted into a text-to-text format.



Training Objective

- Span Corruption (Masked Span Prediction)

Instead of masking individual words, T5 masks continuous spans of text.

Strengths

- Handles both understanding and generation tasks.
- Unified framework for all NLP tasks.

High performance in:

- Translation
 - Summarization
 - Question Answering
 - Dialogue Systems
 - Text Classification
- Excellent transfer learning capability.
 - Simplifies task formulation.

Limitations

- Large computational requirements.
- Slower inference than encoder-only models.
- More memory intensive.
- Complex architecture increases deployment cost.

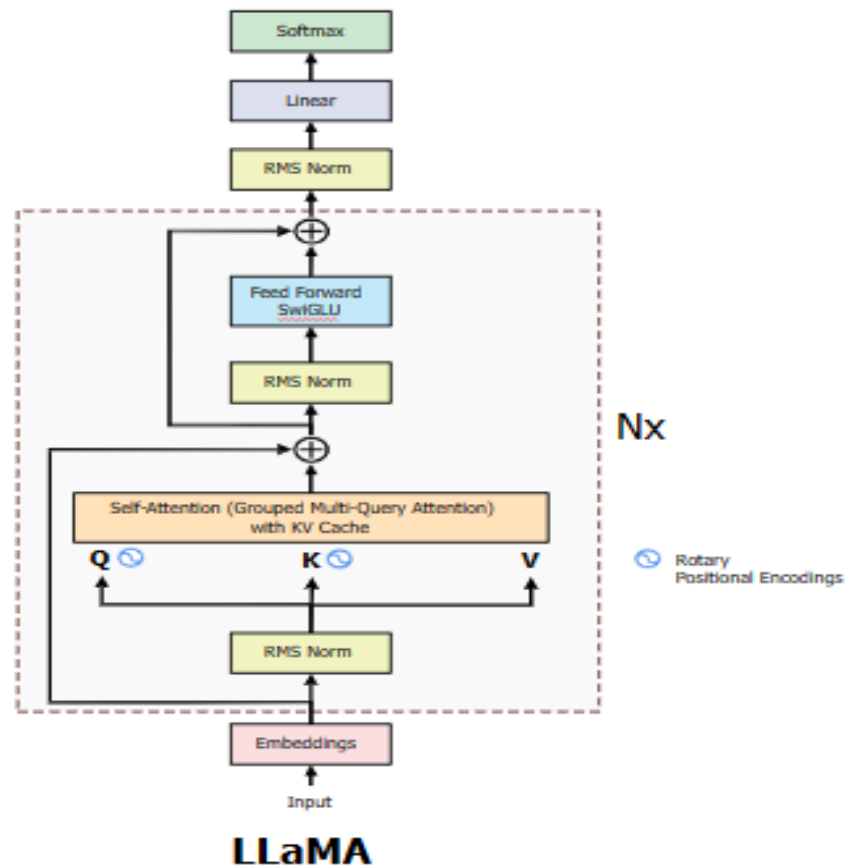
Applications

- Machine translation
- Text summarization
- Conversational AI
- Reading comprehension
- Information extraction

4. LLaMA (Large Language Model Meta AI)

Architecture

LLaMA, developed by Meta AI, is a Transformer Decoder-only model similar to GPT but designed for higher efficiency. It is trained using optimized datasets and modern scaling techniques, enabling strong performance with comparatively fewer parameters.



Training Objective

- Autoregressive Next Token Prediction

The model predicts the next token using previously generated context.

Strengths

- High efficiency compared to many large language models.
- Strong reasoning and coding abilities.
- Open-weight model suitable for research and customization.

- Performs well even with fewer parameters.
- Easier to fine-tune for domain-specific tasks.

Limitations

- Decoder-only architecture limits bidirectional understanding.
- Can generate hallucinated or inaccurate responses.
- Requires alignment and fine-tuning for safe deployment.
- Large models still demand high-end hardware.

Applications

- AI assistants
- Code generation
- Research
- Academic NLP
- Enterprise chatbots
- Document generation

Architectural Comparison

Dimension	BERT	GPT (GPT-3/4 series)	T5 (Text-to-Text Transfer)	LLaMA (LLaMA 1/2/3)
Transformer Backbone	Encoder-Only	Autoregressive Decoder-Only	Encoder-Decoder	Modern Decoder-Only
Attention Mechanism	Bidirectional (Full self-attention across all tokens)	Causal (Unidirectional masking: token t attends only to $\leq t$)	Bidirectional (Encoder) + Causal Cross-Attention (Decoder)	Causal (Grouped-Query Attention / RoPE)
Pre-training Objective	Masked Language Modeling (MLM) + Next Sentence Prediction (NSP)	Causal Next-Token Prediction (Autoregressive LM)	Span Corruption / Denoising Objective	Causal Next-Token Prediction
Positional Encoding	Absolute Learned Positional Embeddings	Absolute Learned Positional Embeddings	Relative Position Biases	Rotary Position Embedding (RoPE)
Normalization / Activation	Post-LN / Pre-LN, GELU	Pre-LN (LayerNorm), GELU	Pre-LN (RMSNorm), Gated GELU / ReGLU	RMSNorm, SwiGLU activation
Primary Sweet Spot	Deep NLU & Representation Extraction	Open-ended NLG & In-Context Learning	Structured Transformation (Translation, Summarization)	Scalable Foundational NLU + High-throughput N

CONCLUSION :-

The BERT architecture uses full bidirectional self-attention and stacked transformer encoder layers to capture rich, left-and-right contextual relationships across text. This design makes it highly effective and robust for downstream natural language understanding, classification, and sequence-tagging tasks.

Question 2:-

A university wants to develop an AI student-support system. Compare BERT, GPT, T5, and LLaMA and recommend the most suitable model for the system.

ANSWER :-

Introduction

Universities are increasingly adopting Artificial Intelligence (AI) to improve student services and provide personalized learning experiences. An AI student-support system can assist students by answering academic queries, explaining course concepts, providing study materials, helping with assignment preparation, scheduling reminders, offering career guidance, and providing information about university policies. To build such a system, selecting the appropriate Natural Language Processing (NLP) model is essential. Among the leading Transformer-based models, BERT, GPT, T5, and LLaMA each offer unique strengths and limitations. This case study compares these models and recommends the most suitable one for a university AI student-support system.

System Requirements

The AI student-support system should be capable of:

- Understanding students' questions accurately.
- Generating clear and natural responses.
- Answering academic and administrative queries.
- Summarizing lecture notes and study materials.
- Translating educational content into multiple languages.
- Providing 24/7 conversational support.
- Maintaining context during long conversations.
- Being cost-effective and scalable.

Comparison of Models

1. BERT

Suitability

BERT is highly effective at understanding the meaning and context of students' questions because it uses a bidirectional encoder architecture. It performs exceptionally well in intent recognition, sentiment analysis, and question classification. For example, BERT can accurately determine whether a student's question is related to admissions, examinations, fee payment, or academic subjects.

Advantages

- Excellent language understanding.
- High accuracy in question classification.
- Strong contextual representation.
- Reliable for information retrieval.

Disadvantages

- Cannot generate detailed conversational responses.
- Requires separate models for different tasks.
- Not suitable as a standalone chatbot.

Use in University

BERT is ideal for classifying student queries and searching university documents but is not sufficient for generating interactive conversations.

2. GPT

Suitability

GPT is designed primarily for natural language generation. It can answer students' questions conversationally, explain difficult concepts, generate study notes, assist with assignments, and provide personalized learning support. GPT can also maintain context throughout a conversation, making interactions more natural.

Advantages

- Produces fluent and human-like responses.
- Supports long conversations.
- Explains academic concepts clearly.
- Generates summaries, emails, reports, and study material.
- Performs well with minimal task-specific training.

Disadvantages

- May occasionally generate incorrect or fabricated information (hallucinations).
- Requires careful monitoring in academic environments.
- Large models demand significant computational resources.

Use in University

GPT is highly suitable as the primary conversational assistant for answering student queries and providing academic guidance.

3. T5

Suitability

T5 is an encoder-decoder model that treats every NLP task as a text-to-text problem. It performs both language understanding and language generation efficiently. T5 can summarize lecture notes, translate educational content, answer questions, rewrite assignments, and generate explanations.

Advantages

- Excellent performance in multiple NLP tasks.
- Strong translation and summarization capabilities.
- Handles both understanding and generation effectively.
- Flexible architecture suitable for educational applications.

Disadvantages

- Computationally expensive.
- Slower than encoder-only or decoder-only models.
- Requires more memory during deployment.

Use in University

T5 is particularly useful for educational content processing, automatic summarization, and multilingual learning support.

4. LLaMA

Suitability

LLaMA is a decoder-only language model that provides strong reasoning, conversational ability, and code generation while being more computationally efficient than many comparable models. Because it is open-weight, universities can fine-tune it using their own academic materials, regulations, and course content.

Advantages

- High efficiency with fewer computational resources.
- Strong reasoning and conversational capabilities.
- Easier to customize using university-specific data.
- Lower deployment cost than many proprietary large language models.
- Suitable for research and on-premises deployment.

Disadvantages

- May still produce inaccurate responses if not properly fine-tuned.
- Requires alignment and safety mechanisms before deployment.
- Language understanding is not as strong as BERT for specialized classification tasks.

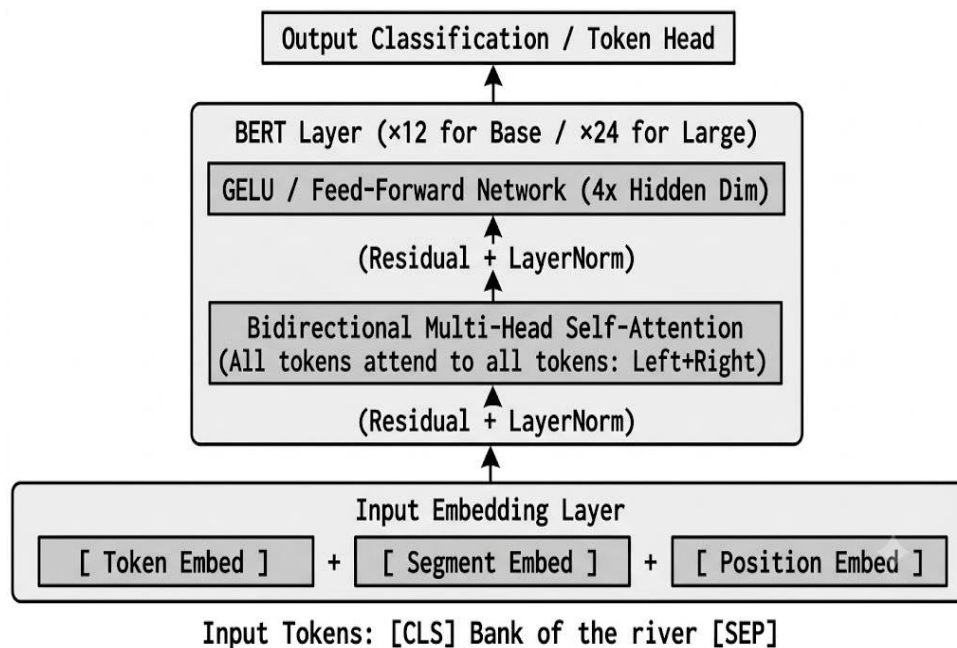
Use in University

LLaMA is an excellent choice for universities that want a customizable AI assistant while maintaining control over their data and infrastructure.

Final Recommendation & Recommended Architecture

The Recommendation: A Hybrid RAG System Powered by LLaMA and BERT

Rather than relying on a single model in isolation, enterprise-grade university deployments succeed through a modular, hybrid architecture:



Why this design wins:

1. BERT for Front-Line Guardrails & Routing:

- Uses a fine-tuned lightweight BERT classifier to instantly route crisis-sensitive topics (mental health, urgent campus safety) directly to human staff without latency.

2. LLaMA (e.g., LLaMA-3.1-8B-Instruct or 70B) for Generative Support:

- Synthesizes retrieved university documentation into concise, student-friendly responses using Retrieval-Augmented Generation (RAG).

3. FERPA & Privacy Guarantee:

- All components run within the university's private subnet, ensuring absolute compliance with educational privacy laws and protecting student personal information.

Comparison Table

Feature	BERT	GPT	T5	LLaMA
Architecture	Encoder-only	Decoder-only	Encoder-Decoder	Decoder-only
Language Understanding	Excellent	Very Good	Excellent	Very Good
Conversational Ability	Poor	Excellent	Excellent	Excellent
Text Generation	Poor	Excellent	Excellent	Excellent
Question Answering	Excellent	Excellent	Excellent	Very Good
Summarization	Moderate	Excellent	Excellent	Excellent
Translation	Moderate	Good	Excellent	Very Good
Customization	Moderate	Good	Moderate	Excellent
Deployment Cost	Medium	High	High	Moderate
Best Role	Query Classification	AI Chatbot	Multi-task NLP	Custom AI Assistant

Recommendation

For a university AI student-support system, LLaMA is the most suitable choice when the university wants to deploy and customize its own AI assistant using internal academic data while maintaining lower infrastructure costs. Its open-weight architecture allows fine-tuning with university regulations, course materials, and student handbooks, making it highly adaptable.

If the university prefers a cloud-based solution with the strongest conversational capabilities and does not require hosting the model itself, GPT is an excellent alternative because it provides natural, context-aware responses and can assist students with explanations, writing support, and interactive learning.

For the best overall system, a hybrid approach is recommended:

- BERT for classifying and retrieving relevant university information.
- GPT or LLaMA for conversational responses and student interaction.
- T5 for summarizing lecture notes, translating educational content, and rewriting study materials.

This combination provides accurate understanding, high-quality response generation, multilingual support, and efficient handling of diverse academic tasks.

Conclusion

Each model has distinct strengths that make it suitable for different aspects of a university AI student-support system. BERT excels in language understanding and query classification, GPT provides highly fluent conversational responses, T5 is ideal for text transformation tasks such as summarization and translation, and LLaMA offers an efficient, customizable solution for institutions that want greater control over deployment. For most universities, a hybrid architecture combining BERT with GPT or LLaMA, while using T5 for specialized text-processing tasks, delivers the most accurate, scalable, and user-friendly AI student-support system.